Nabila Alia Zahra	**Rangkuman Minggu 10** **Pengaplikasian Arduino menggunakan Tinkercad**	Senin, 07 November 2022
		16TNK5043 (Aplikasi Mikrokontroler)
3NK		Ferry Satria, BSEE.,M.T. Rahmawati Hasanah,S.ST.,M.T.

Arduino :

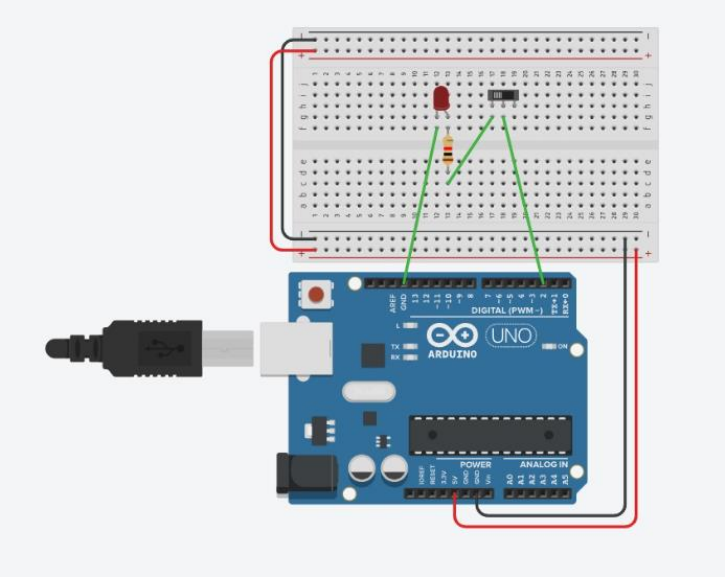
Void Setup → untuk inisialisasi dan deklarasi

Void Loop → untuk memasukan main program

Program untuk mengaktifkan LED, logic 1 menyala, logic 0 mati

```
// C++ code
//
void setup()
{
  pinMode(2, OUTPUT);
}

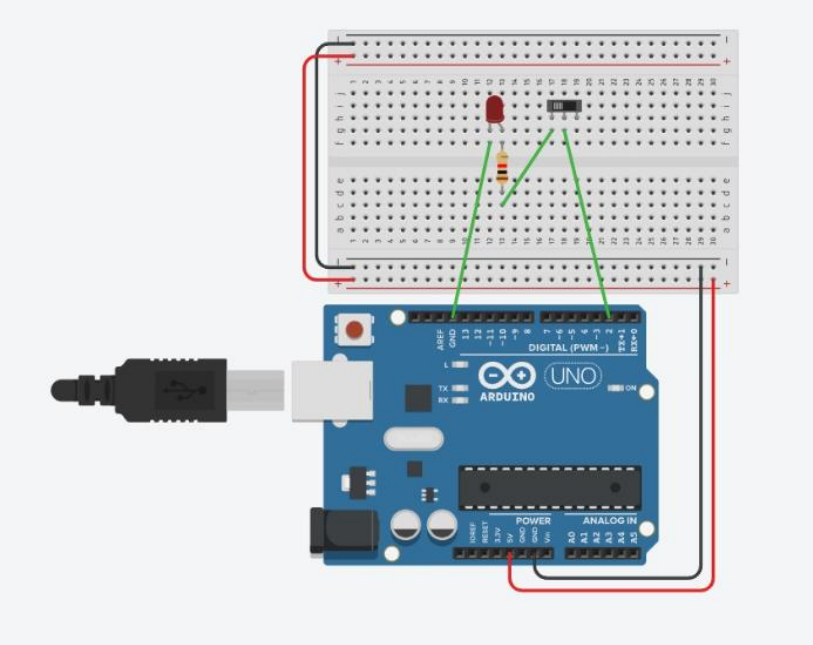
void loop()
{
  digitalWrite(2, HIGH);
}
```



Program untuk mengaktifkan LED, logic 1 flip flop, logic 0 mati

```
// C++ code
//
void setup()// inisialisasi dan deklarasi
{
  pinMode(2, OUTPUT);//Supaya jadi output
}

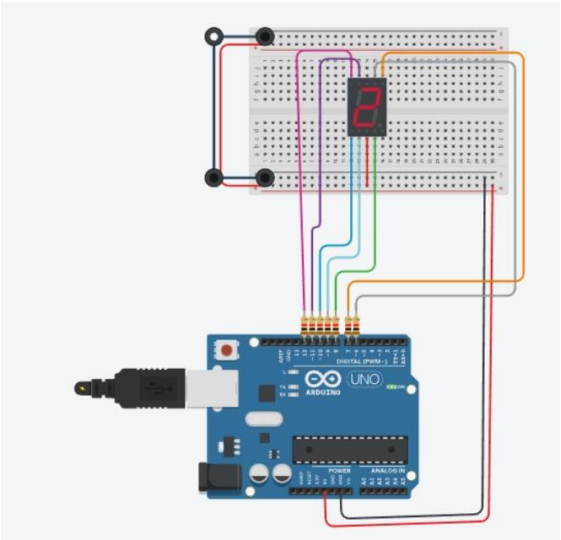
void loop()//Main program
{
  digitalWrite(2, HIGH);// biar berlogic 1
  delay(1000); // Wait for 1000 millisecond(s)
  digitalWrite(2, LOW);
  delay(1000); // Wait for 1000 millisecond(s)
}
```



Program untuk menampilkan angka 2 pada seven segmen common cathode

```
int a = 6;    //Segment A
int b = 7;    //Segment B
int c = 8;    //Segment C
int d = 9;    //Segment D
int e = 10;   //Segment E
int f = 11;   //Segment F
int g = 12;   //Segment G
int k,n;
int pin[] = {0,0,1,0,0,1,0};
void setup()
{
  for (k=6 ; k<13 ; k++)
    pinMode (k,OUTPUT);
}
```

```
void loop() {
  for (n=0 ; n<7 ; n++) {
    digitalWrite(n+6 , pin[n]);
    }
}
```

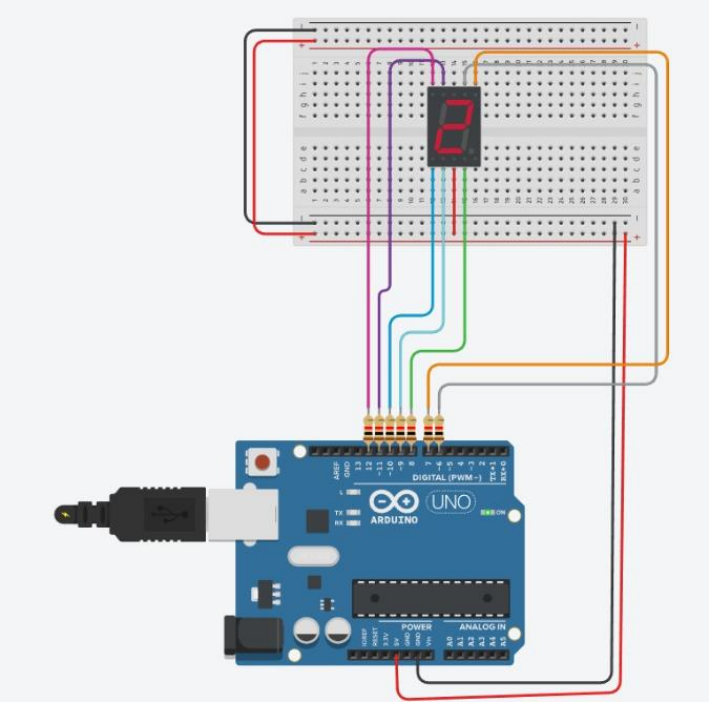


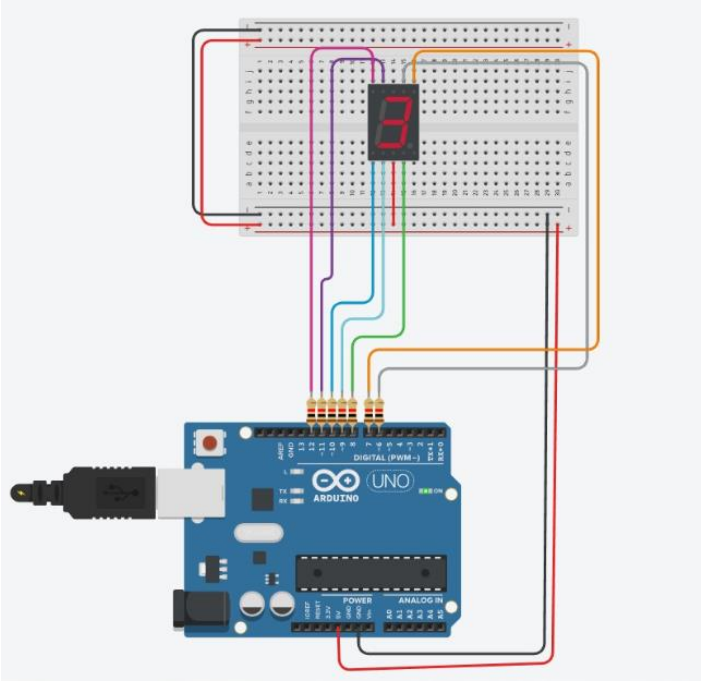
Program menampilkan huruf 2 dan 3 secara bergantian

```
int a = 6;    //Segment A
int b = 7;    //Segment B
int c = 8;    //Segment C
int d = 9;    //Segment D
int e = 10;   //Segment E
int f = 11;   //Segment F
int g = 12;   //Segment G
int k,n;
int pin1[] = {0,0,1,0,0,1,0};
int pin2[] = {0,0,0,0,1,1,0};
void setup()
{
  for (k=6 ; k<13 ; k++)
    pinMode (k,OUTPUT);
}
void loop() {
  for (n=0 ; n<7 ; n++) {
    digitalWrite(n+6 , pin1[n]);
    }
  delay(1000);
  for (n=0 ; n<7 ; n++) {
    digitalWrite(n+6 , pin2[n]);
    }
  delay(1000);
}
```

```
int a = 6;    //Segment A
int b = 7;    //Segment B
int c = 8;    //Segment C
int d = 9;    //Segment D
int e = 10;   //Segment E
int f = 11;   //Segment F
int g = 12;   //Segment G
int k,n;
int pin1[] = {0,0,0,0,0,0,1};
int pin2[] = {1,0,0,1,1,1,1};
int pin3[] = {0,0,1,0,0,1,0};
int pin4[] = {0,0,0,0,1,1,0};
int pin5[] = {1,0,0,1,1,0,0};
int pin6[] = {0,1,0,0,1,0,0};
int pin7[] = {0,1,0,0,0,0,0};
int pin8[] = {0,0,0,1,1,1,1};
int pin9[] = {0,0,0,0,0,0,0};
int pin10[] = {0,0,0,0,1,0,0};
void setup()
{
  for (k=6 ; k<13 ; k++)
    pinMode (k,OUTPUT);
}
void loop() {
  for (n=0 ; n<9 ; n++) {
    digitalWrite(n+6 , pin1[n]);
    }
  delay(1000);
  for (n=0 ; n<9 ; n++) {
    digitalWrite(n+6 , pin2[n]);
    }
  delay(1000);
  for (n=0 ; n<9 ; n++) {
    digitalWrite(n+6 , pin3[n]);
    }
  delay(1000);
  for (n=0 ; n<9 ; n++) {
    digitalWrite(n+6 , pin4[n]);
    }
  delay(1000);
  for (n=0 ; n<9 ; n++) {
    digitalWrite(n+6 , pin5[n]);
    }
  delay(1000);
  for (n=0 ; n<9 ; n++) {
    digitalWrite(n+6 , pin6[n]);
    }
  delay(1000);
  for (n=0 ; n<9 ; n++) {
    digitalWrite(n+6 , pin7[n]);
```

```
    }
  delay(1000);
  for (n=0 ; n<9 ; n++) {
    digitalWrite(n+6 , pin8[n]);
    }
  delay(1000);
  for (n=0 ; n<9 ; n++) {
    digitalWrite(n+6 , pin9[n]);
    }
  delay(1000);
  for (n=0 ; n<9 ; n++) {
    digitalWrite(n+6 , pin10[n]);
    }
  delay(1000);
}
```




DIGITAL (PWM~)
UNO
ARDUINO
POWER
ANALOG IN